Chulalongkorn University · Department of Chemical Engineering

Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator

Instrument list and I/O schedule

NDFH-V1-IN-INL-001 · Rev 0 · Issued for teaching

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Title block	
Client	Chulalongkorn University · Department of Chemical Engineering
Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator
Document no.	NDFH-V1-IN-INL-001
Title	Instrument list and I/O schedule
Drawings	PFD-100 rev 1; P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev E (EPC alternative)
Scope	Every instrument and final element on the P&ID with range, settings, I/O and fail position; I/O count; checks.

All numbers in this document come from the Natural Draft Fired Heater Ver 1 simulator model (design case and model runs) or are stated as assumed. It is a teaching document, not a design for construction.

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1 · Instrument list

Tag	Service	Type	System	Range	Unit	Normal	Alarm	Trip / action	I/O	Fail	P&ID
FT-008	Feed to H-100 (FE-008)	Flow transmitter, orifice	DCS + SIS	0 … 12	L/s	6.80	L 5	FSLL-008 L 3.4	AI	—	PID-001
FIC-003	Feed flow controller	DCS function	DCS	0 … 12	L/s	6.80	—	—	—	—	PID-001
FV-010	Feed flow control valve	Globe control valve	DCS	0 … 100	%	56.67	—	—	AO	FO	PID-001
PI-001	P-100 suction	Pressure gauge	Field	-100 … 400	kPa(g)	50	—	—	—	—	PID-001
PI-002	P-100 discharge	Pressure gauge	Field	0 … 2000	kPa(g)	1030	—	—	—	—	PID-001
PI-009	Coil inlet	Pressure transmitter	DCS	0 … 2000	kPa(g)	845	—	—	AI	—	PID-001
PI-005	Heater outlet	Pressure transmitter	DCS	0 … 1000	kPa(g)	545	—	—	AI	—	PID-001
TI-010	Feed after E-100	Temperature transmitter	DCS	0 … 300	°C	180.0	L 170	—	AI	—	PID-001
TI-063	Heating fluid from Unit A	Temperature transmitter	DCS	0 … 300	°C	230.0	—	—	AI	—	PID-001
HV-063	Heating fluid to E-100	Hand control valve	DCS	0 … 100	%	60.00	—	—	AO	FL	PID-001
TI-020	Coil inlet temperature	Temperature transmitter	DCS	0 … 300	°C	180.3	—	—	AI	—	PID-001
TI-022	Crossover temperature	Temperature transmitter	DCS	0 … 400	°C	233.3	—	—	AI	—	PID-001
TT-040	Heater outlet (control)	Temperature transmitter, thermocouple	DCS	0 … 500	°C	350.0	L 344 / H 356	—	AI	—	PID-001
TIC-030	Outlet temperature controller	DCS function	DCS	0 … 500	°C	350.0	—	—	—	—	PID-001
TE-025	Heater outlet (independent)	Temperature transmitter, own thermowell	SIS	0 … 500	°C	351.3	H 360	TSHH-025 H 375	AI	—	PID-001
TI-024	Radiant tube skin	Skin thermocouple	DCS	0 … 700	°C	397.9	H 430	—	AI	—	PID-001
TIAH-023	Firebox (bridgewall)	Temperature transmitter	DCS	0 … 1400	°C	986.2	H 1100	—	AI	—	PID-001
TI-021	Stack	Temperature transmitter	DCS	0 … 600	°C	399.9	H 470	—	AI	—	PID-001
PI-003	Firebox pressure (arch)	Draft transmitter	DCS	-10 … 5	mmH₂O	−2.50	H -0.5	—	AI	—	PID-001
PI-004	Firebox pressure (floor)	Draft transmitter	DCS	-20 … 5	mmH₂O	−5.97	—	—	AI	—	PID-001
PI-031	Convection outlet draft	Draft transmitter	DCS	-10 … 5	mmH₂O	−1.25	—	—	AI	—	PID-001
PI-032	Draft above damper	Draft transmitter	DCS	-10 … 5	mmH₂O	−2.93	—	—	AI	—	PID-001
PT-030	Windbox draft	Draft transmitter	DCS + SIS	0 … 10	mmH₂O	5.25	L 1.5	PSLL-030 L 0.5	AI	—	PID-001
FT-005	Combustion air (FE-005)	Flow transmitter, venturi	DCS	0 … 6000	kg/h	3892	—	—	AI	—	PID-001
FIC-005	Combustion air controller	DCS function	DCS	0 … 6000	kg/h	3892	—	—	—	—	PID-001
FV-200	Combustion air valve	Butterfly control valve	DCS	0 … 100	%	47.52	—	—	AO	FO	PID-001
HV-400	Stack damper	Damper actuator	DCS	0 … 100	%	67.00	—	—	AO	FO	PID-001
FI-001	Fuel oil (FE-001)	Flow transmitter	DCS	0 … 400	kg/h	243	—	—	AI	—	PID-001
FV-300	Fuel oil control valve	Globe control valve	DCS	0 … 100	%	45.32	—	—	AO	FC	PID-001
PI-019	Fuel oil header	Pressure transmitter	DCS	0 … 3000	kPa(g)	1940	—	—	AI	—	PID-001
PT-020	Fuel oil at the gun	Pressure transmitter	DCS + SIS	0 … 2000	kPa(g)	600	L 100 / H 1400	PSLL-020 L 40	AI	—	PID-001
XV-301	Fuel oil block valve 1	On-off valve, solenoid	SIS	—	—	—	—	closes on trip	DO + 2 DI	FC	PID-001
XV-302	Fuel oil block valve 2	On-off valve, solenoid	SIS	—	—	—	—	closes on trip	DO + 2 DI	FC	PID-001
XV-303	Fuel oil vent between blocks	On-off valve, solenoid	SIS	—	—	—	—	opens on trip	DO + 2 DI	FO	PID-001
BS-005	Flame scanner	UV/IR flame scanner	SIS	—	—	1	—	BSLL-005 flame lost	DI	—	PID-001

Tag	Service	Type	System	Range	Unit	Normal	Alarm	Trip / action	I/O	Fail	P&ID
IGN-101	Igniter	Ignition transformer	SIS	—	—	0	—	off on trip	DO	—	PID-001
AT-028	Firebox O_2	Zirconia analyzer	DCS	0 … 10	% O_2	2.10	L 1 / H 6	—	AI	—	PID-001
AT-029	Stack O_2	Zirconia analyzer	DCS	0 … 10	% O_2	3.22	L 1.5 / H 7	—	AI	—	PID-001
AT-026	Flue CO	Analyzer (TDL)	DCS	0 … 1000	ppm	20	H 200	—	AI	—	PID-001
HS-001	Emergency trip push-button	Hand switch, control room	SIS	—	—	—	—	ESD	DI	—	PID-001
HS-221	Snuffing steam HV-221	Manual valve with limit switch	DCS	—	—	0	—	purge permissive	DI	—	PID-001

2 · I/O schedule

Wired to	AI	AO	DI	DO	Total
DCS	20	5	1	0	26
SIS	1	0	8	4	13
DCS + SIS	3	0	0	0	3
DCS incl. shared	23	5	1	0	29
SIS incl. shared	4	0	8	4	16
DCS + 20 % spare	28	6	2	0	35
SIS + 20 % spare	5	0	10	5	20

3 · Fail positions and the trip action

Final element	Fail position	What the trip does (simulator)	Same?
XV-301	FC (de-energise)	trip: closed	✓
XV-302	FC (de-energise)	trip: closed	✓
XV-303	FO (de-energise)	trip: vent open	✓
FV-300	FC	trip: TIC-030 MAN 0 % → closed	✓
FV-200	FO	trip: FIC-005 MAN 100 % → open	✓
HV-400	FO	trip: 100 % open (sweep the box)	✓
FV-010	FO	trip: feed keeps flowing (coil cooled)	✓

Cause and effect: NDFH-V1-IN-CE-001 (C&E matrix tab).

4 · Checks

Check	This document	Compared with	Result
Alarm FIAL-008 inside the range of FT-008	0 … 12 L/s	5 (ALM)	✓
Alarm TIC-030 inside the range of TT-040	0 … 500 °C	344 / 356 (ALM)	✓
Alarm TIAH-025 inside the range of TE-025	0 … 500 °C	360 (ALM)	✓
Alarm TIAH-023 inside the range of TIAH-023	0 … 1400 °C	1100 (ALM)	✓
Alarm AI-026 inside the range of AT-026	0 … 1000 ppm	200 (ALM)	✓
Alarm AI-028 inside the range of AT-028	0 … 10 % O_2	1 / 6 (ALM)	✓
Alarm AI-029 inside the range of AT-029	0 … 10 % O_2	1.5 / 7 (ALM)	✓
Alarm PI-003 inside the range of PI-003	-10 … 5 mmH_2O	-0.5 (ALM)	✓
Alarm PIAL-020 inside the range of PT-020	0 … 2000 kPa(g)	100 / 1400 (ALM)	✓
Alarm PIAL-030 inside the range of PT-030	0 … 10 mmH_2O	1.5 (ALM)	✓
Alarm TI-021 inside the range of TI-021	0 … 600 °C	470 (ALM)	✓
Alarm TI-024 inside the range of TI-024	0 … 700 °C	430 (ALM)	✓
Alarm TI-010 inside the range of TI-010	0 … 300 °C	170 (ALM)	✓
Trip FSLL-008 setting inside the range of FT-008 and = simulator	FSLL-008 L 3.4	3.4 (K)	✓
Trip TSHH-025 setting inside the range of TE-025 and = simulator	TSHH-025 H 375	375 (K)	✓
Trip PSLL-020 setting inside the range of PT-020 and = simulator	PSLL-020 L 40	40 (K)	✓
Trip PSLL-030 setting inside the range of PT-030 and = simulator	PSLL-030 L 0.5	0.5 (K)	✓
Normal value 10–90 % of span (pressure, temperature and flow transmitters and gauges)	22 instruments	all inside (see the range bar in section 2)	✓
Every P&ID bubble on the DCS drawing is in the instrument list	41 rows	all 27 found	✓
FSLL-008 has its own transmitter (independent of FIC-003)	FT-008 feeds FIC-003, FIAL-008 and FSLL-008	Safety layers tab, independence check	✗
TSHH-025 independent of the control loop	TE-025 own thermowell	TT-040 for TIC-030	✓
XV-301 fail position = trip action	FC (de-energise)	trip: closed	✓
XV-302 fail position = trip action	FC (de-energise)	trip: closed	✓
XV-303 fail position = trip action	FO (de-energise)	trip: vent open	✓
FV-300 fail position = trip action	FC	trip: TIC-030 MAN 0 % → closed	✓
FV-200 fail position = trip action	FO	trip: FIC-005 MAN 100 % → open	✓
HV-400 fail position = trip action	FO	trip: 100 % open (sweep the box)	✓
FV-010 fail position = trip action	FO	trip: feed keeps flowing (coil cooled)	✓